AI BASED QUESTION BANK PREDICTION SYSTEM BASED ON NCERT CURRICULUM

SYNOPSIS REPORT

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ABSTRACT

AI based question bank prediction system based on Ncert curriculum is a web-based AI-enabled educational platform designed to help students practice class-wise and chapter-wise questions based class-wise, subject-wise, and chapter-wise categorization. The platform provides instant evaluation of student responses and displays detailed performance analytics including correct/wrong count, score, and chapter mastery percentage.

The system is optimized for use on tablets through an "Add to Home Screen" feature, allowing students to use the platform as a clean, distraction-free app without installation. The backend integrates AI-based question generation, auto-evaluation logic, and progress tracking, while the frontend is built using HTML, CSS, and JavaScript.

Node.js is used for server-side processing, and Firebase/MySQL is used for data storage. AI Study Point aims to enhance self-learning, strengthen exam preparation, and make NCERT practice more interactive, personalized, and accessible. The platform ultimately supports students with smart AI-generated questions, real-time feedback, and consistent learning progress.

1. INTRODUCTION

AI based question bank prediction system based on Ncert curriculum is a web-based AI-powered learning platform designed specifically to address these challenges. The platform provides class-wise, subject-wise, and chapter-wise question practice for NCERT Classes 6 to 8. Unlike static question banks, AI based question bank prediction system based on Ncert curriculum uses artificial intelligence to automatically read NCERT chapters and generate multiple question types including MCQs, True/False, and short-answer questions.

This ensures that learners always receive updated, chapter-relevant, and concept-focused practice. Digital learning is becoming essential for students in India, especially those using NCERT textbooks. Students often struggle to find structured, chapter-wise questions with instant explanations. The system focuses on NCERT textbooks. The platform uses artificial intelligence to automatically generate MCQ, True/False, and short-answer questions from NCERT chapters and evaluates students' answers instantly. students can practice these questions through a structured interface that includes

on simplicity and accessibility, especially for Class 6–8 students who use tablets in schools. Through the "Add to Home Screen" feature, the website works like a mobile/tablet app without installation.

1.1 Motivation and Significance

A major motivation behind AI based question bank prediction system based on Ncert curriculum is to remove these learning barriers and provide students with a reliable, intelligent, and easy-to-use practice system. Many students struggle to find chapter-specific questions, practice regularly, and understand their weak areas. Teachers also require platforms that can assist with revision, homework, and concept reinforcement.The significance of this project lies in its ability to transform NCERT-based learning into a personalized and interactive experience. Its tablet-friendly design makes it ideal for modern school environments, where digital devices are increasingly used for education. By providing quick access, clean UI, and AI-powered evaluation, AI based question bank prediction system based on Ncert curriculum enhances learning efficiency and supports both home-based study and classroom learning. Students need Organized question practice, Instant evaluation, Progressive difficulty levels, Chapter mastery indicators, AI based question bank prediction system based on Ncert curriculum directly fulfills these needs with AI-powered question generation and instant performance analytics.

2. PROBLEM STATEMENT AND OBJECTIVES

Students who rely on NCERT textbooks often face multiple challenges while preparing for exams, especially when it comes to chapter-wise practice and self-assessment. Most available learning platforms provide either unstructured question banks, paid content, or lack AI-driven evaluation. Students find it difficult to:

2.1 Core Problem

Students need a centralized AI-powered system for structured chapter practice.

- Unstructured or outdated questions
- No AI-driven instant evaluation
- Lack of chapter-wise progression
- No unified free platform for Class 6–8
- No device-friendly interface for school tablets

2.2 Project Objectives

These objectives collectively aim to create an intelligent, user-friendly platform that enhances NCERT learning through AI-driven automation and personalized practice.

The main objectives of the AI Study Point project are:

- To design a web-based NCERT practice system named AI based question bank prediction system based on Ncert curriculum. To enable class-wise, subject-wise, and chapter-wise navigation. To design a clean, simple, and tablet-optimized interface that works like an app through the "Add to Home Screen" feature.
- To develop AI-generated questions (MCQ, True/False, Short Answer). To develop an admin interface for managing questions, student data, and system content when required. that makes navigation easy for Class 6–8 students and supports effective self-learning.
- To implement automated evaluation of student responses. To integrate an AI-powered question generation system. that extracts content from NCERT chapters and automatically creates MCQs, True/False, and short-answer questions.
- To create a tablet-friendly, distraction-free UI.

3. SCOPE AND METHODOLOGY

The scope of AI based question bank prediction system based on Ncert curriculum encompasses the development of an AI-powered, web-based learning platform designed to support NCERT-based practice for students of Classes 6 to 8. The project focuses on creating a structured and intelligent environment where students can learn, practice, and assess their chapter-wise understanding through automatically generated questions and real-time evaluation.

3.1 Inclusions

- Class -wise and Subject-wise Navigation:-The system provides structured access to NCERT content for Classes 6, 7, and 8 across key subjects such as Mathematics, Science, Social Science, English, and Hindi.
- AI-Based Question Generation:- The project includes the integration of AI/NLP techniques to read NCERT chapters and automatically generate various question types: Multiple Choice Questions (MCQs), True/False, Short-answer questions, Difficulty-based questions (Easy, Medium, Hard)
- Chapter-WisePractice and Quiz Interface:-Students can start quizzes for any selected chapter and practice questions interactively.

- Instant Evaluation and Analytics:-The system provides real-time scoring, answer feedback, correct/wrong count, and chapter mastery percentage.
- Student Progress Tracking:-The platform stores score history, attempted quizzes, performance trends, and overall progress.
- AdminPanel(Optional):-Admins can upload, modify, or delete questions and review student performance logs.
- Tablet-Friendly UserInterface:-The system supports "Add to Home Screen" functionality, enabling app-like usage, full-screen mode, and distraction-free learning—ideal for school tablets.
- Secure Scalable Web Deployment:-The project includes hosting setup, SSL integration, and responsive design for smooth use across devices.

3. 2 Exclusions

- Offline standalone mobile applications (Android/iOS).
- Non-NCERT syllabus such as CBSE reference books, state boards, or competitive exam books.
- Video lectures, animations, or live teaching modules.
- Advanced AI models for long-form descriptive answer checking.
- Integration with school ERP or attendance systems.

4. EXISTING

The existing system for NCERT-based question practice is largely traditional and lacks automation, personalization, and structured digital accessibility. Students typically rely on physical textbooks, handwritten notes, and manual question banks to revise chapter-wise concepts. This method is time-consuming and does not provide instant evaluation or performance tracking. There are some online educational platforms available, but they also come with limitations:

1. Manual NCERT Practice

- Students refer to books or PDFs for questions.
- Answers must be checked manually.
- No analytics or chapter mastery indicators.

2. Online Quiz Websites

- Provide general quizzes but not NCERT-specific content.

- Questions are manually added and not generated dynamically.
- Lack class-wise and chapter-wise navigation.

3. Educational Apps

- Many apps offer NCERT solutions,
- They are mostly paid or subscription-based.
- They do not offer AI-generated questions.
- They lack structured difficulty levels (Easy/Medium/Hard).
- Instant feedback and personalized progress tracking are limited.

4. School-Based Digital Learning

- Some schools use tablets or smart classes,
- Resources are not interactive.
- No AI-powered adaptive practice.
- No centralized question practice system.

5. PROPOSED METHODOLOGY

- The proposed methodology for AI based question bank prediction system based on Ncert curriculum focuses on building an AI-enabled, web-based NCERT learning platform that allows students to practice chapter-wise questions with instant evaluation and progress tracking. The methodology is divided into systematic phases that ensure smooth development, AI integration, and effective learning outcomes.

System Flow

- Home Page → Class Selection
- Subjects → Chapter List
- AI Model → Reads NCERT chapter
- Generates questions automatically
- Student answers → System evaluates
- Generates result + mastery percentage

Backend Logic

AI Model Performs:

- Text extraction from NCERT PDFs

- Auto-generation of question types
- Difficulty categorization
- Answer matching & scoring

Database Stores:

- Student details Question sets
- Scores & progress logs
- Mastery percentages

Implementation Plan

- UI design & structure
- Backend & AI setup
- Question generation module
- Testing & validation
- Deployment + tablet integration

6. EXPECTED OUTCOME

The development of AI based question bank prediction system based on Ncert curriculum is expected to produce an intelligent, user-friendly, and efficient learning system that enhances NCERT-based practice for students. The anticipated outcomes of the project are as follows:

1.AI-Generated NCERT Question Bank

The system will automatically generate chapter-wise questions including:

- Multiple Choice Questions (MCQs)True/False
- Short answer questions
- Difficulty levels (Easy / Medium / Hard)

2. Automated Evaluation System Students will receive:

This ensures relevant, updated, and concept-based practice.

- Instant scoring
- Correct/wrong feedback
- Accuracy percentage
- Chapter mastery percentage

3. Personalized Learning Experience The platform will provide: This promotes fast learning and accurate self-assessment.

- Progress tracking
- Attempt history
- Weak/strong chapter identification
- Adaptive difficulty practice

4. Interactive and Structured Learning Interface

This helps students improve performance gradually.

- Class-wise and subject-wise organized navigation
- Clean and child-friendly UI
- One-question-per-screen quiz format
- Chapter progress indicators

5. Tablet-Friendly App-Like System

By using the "Add to Home Screen" feature, the system will:

- Work like a mobile/tablet app
- Run in full-screen mode
- Provide a distraction-free environment
- Be ideal for school tablet setups

6. Admin Panel for Content Management

Optional admin features will allow:

- Addition, modification, and deletion of questions
- Monitoring of student activity and performance
- Database updates and chapter management

7. Secure, Scalable, and Deployable Web System

The final product will:

- Be fully hosted on a secure domain
- Support SSL-based HTTPS access
- Handle multiple users simultaneously
- Be accessible on laptops, mobiles, and tablets

7. TOOLS AND TECHNOLOGIES USED

Frontend:
HTML, CSS, JavaScript

Backend:
Node.js / Express.js

Database:
Firebase / MySQL

AI Technologies:
NLP-based model
Text extraction (OCR/PDF Reader)
Question generation algorithms

Others:
Git, VS Code
Web hosting + SSL
Add to Home Screen (PWA behavior)

8.CONCLUSION

The development of AI based question bank prediction system based on Ncert curriculum demonstrates how artificial intelligence can significantly enhance the learning experience for students studying NCERT-based subjects. The platform provides a structured, interactive, and efficient method for practicing chapter-wise questions for Classes 6 to 8. By integrating AI-driven question generation, instant evaluation, and progress tracking, the system addresses the limitations of traditional learning methods and existing educational platforms.

AI based question bank prediction system based on Ncert curriculum not only simplifies access to NCERT practice material but also personalizes the learning process by offering chapter mastery analysis, difficulty-level quizzes, and performance monitoring. The clean, user-friendly interface and tablet-optimized design ensure that students can use the system easily at home or in school digital classrooms. The “Add to Home Screen” feature further transforms the web platform into an app-like environment, supporting distraction-free learning.

This project highlights the potential of technology and AI in improving educational outcomes by making practice more engaging, adaptive, and accessible. With continuous enhancements and scalability, AI based question bank prediction system based on Ncert curriculum can evolve into a comprehensive learning ecosystem that supports students, teachers, and schools in achieving better academic performance.

9. REFERENCES

- NCERT Official Website https://ncert.nic.in/.
- National Digital Lirary of India (NDLI) https://ndl.iitkgp.ac.in/.
- VS Code Documentation https://code.visualstudio.com/docs
- MySQL Developer Documentation. https://dev.mysql.com/doc/.
- Node.js & Express Documentation https://nodejs.org/en.
- OpenAI NLP Concepts https://platform.openai.com/docs/overview.
- Git & GitHub Documentation https://docs.github.com/.
- PDF Text Extraction & NLP Concepts https://www.geeksforgeeks.org/nlp-tutorial.